EXPERIMENT 3

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Implement a training algorithm using stochastic gradient descent (SGD) with momentum and compare it with the Adam optimizer. Train both models on a dataset and compare their convergence rates and performance.

```
import tensorflow as tf
import matplotlib.pyplot as plt

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

x_train = x_train / 255.0
x_test = x_test / 255.0

x_train = x_train[:20000]
y_train = y_train[:20000]

def create_model():
    model = tf.keras.Sequential([
        tf.keras.layers.Flatten(input_shape=(32, 32, 3)),
        tf.keras.layers.Dense(256, activation="relu"),
        tf.keras.layers.Dense(128, activation="relu"),
        tf.keras.layers.Dense(10, activation="softmax")
    ])
    return model

model_sgd = create_model()

sgd = tf.keras.optimizers.SGD(
    learning_rate=0.01,
    momentum=0.9
)

model_sgd.compile(
    optimizer=sgd,
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

history_sgd = model_sgd.fit(
    x_train,
    y_train,
    epochs=10,
    batch_size=64,
    validation_split=0.1
)

model_adam = create_model()

adam = tf.keras.optimizers.Adam(
    learning_rate=0.001
)

model_adam.compile(
    optimizer=adam,
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

history_adam = model_adam.fit(
    x_train,
    y_train,
    epochs=10,
    batch_size=64,
    validation_split=0.1
)

loss1, acc1 = model_sgd.evaluate(x_test, y_test)
loss2, acc2 = model_adam.evaluate(x_test, y_test)

print("SGD Accuracy:", acc1)
print("Adam Accuracy:", acc2)

plt.plot(history_sgd.history["loss"], label="SGD + Momentum")
plt.plot(history_adam.history["loss"], label="Adam")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("SGD Momentum vs Adam")
plt.legend()
```

```
plt.show()
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 1566s 9us/step
/usr/local/lib/python3.13/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input_shape
  super().__init__(**kwargs)
Epoch 1/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 16ms/step - accuracy: 0.2845 - loss: 1.9644 - val_accuracy: 0.3100 - val_loss: 1.8571
Epoch 2/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 16ms/step - accuracy: 0.3551 - loss: 1.7872 - val_accuracy: 0.3750 - val_loss: 1.7355
Epoch 3/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 4s 14ms/step - accuracy: 0.3856 - loss: 1.7111 - val_accuracy: 0.3625 - val_loss: 1.7192
Epoch 4/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - accuracy: 0.4091 - loss: 1.6560 - val_accuracy: 0.4160 - val_loss: 1.6289
Epoch 5/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - accuracy: 0.4162 - loss: 1.6305 - val_accuracy: 0.4145 - val_loss: 1.6317
Epoch 6/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 7s 16ms/step - accuracy: 0.4300 - loss: 1.5878 - val_accuracy: 0.3905 - val_loss: 1.6861
Epoch 7/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - accuracy: 0.4497 - loss: 1.5442 - val_accuracy: 0.4300 - val_loss: 1.5788
Epoch 8/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - accuracy: 0.4592 - loss: 1.5152 - val_accuracy: 0.4240 - val_loss: 1.6042
Epoch 9/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 3s 11ms/step - accuracy: 0.4634 - loss: 1.4875 - val_accuracy: 0.4390 - val_loss: 1.5561
Epoch 10/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 4s 15ms/step - accuracy: 0.4783 - loss: 1.4644 - val_accuracy: 0.4405 - val_loss: 1.5920
Epoch 1/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 14ms/step - accuracy: 0.2683 - loss: 2.0184 - val_accuracy: 0.3060 - val_loss: 1.9295
Epoch 2/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.3457 - loss: 1.8304 - val_accuracy: 0.3250 - val_loss: 1.8809
Epoch 3/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 4s 15ms/step - accuracy: 0.3782 - loss: 1.7442 - val_accuracy: 0.3815 - val_loss: 1.7195
Epoch 4/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 14ms/step - accuracy: 0.4004 - loss: 1.6796 - val_accuracy: 0.3840 - val_loss: 1.7241
Epoch 5/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 18ms/step - accuracy: 0.4148 - loss: 1.6528 - val_accuracy: 0.3710 - val_loss: 1.7122
Epoch 6/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 4s 15ms/step - accuracy: 0.4278 - loss: 1.6063 - val_accuracy: 0.4025 - val_loss: 1.6464
Epoch 7/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 14ms/step - accuracy: 0.4459 - loss: 1.5602 - val_accuracy: 0.4195 - val_loss: 1.6264
Epoch 8/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 19ms/step - accuracy: 0.4567 - loss: 1.5260 - val_accuracy: 0.4190 - val_loss: 1.6383
Epoch 9/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 4s 14ms/step - accuracy: 0.4586 - loss: 1.5091 - val_accuracy: 0.4235 - val_loss: 1.5856
Epoch 10/10
282/282 ━━━━━━━━━━━━━━━━━━━━ 5s 14ms/step - accuracy: 0.4710 - loss: 1.4821 - val_accuracy: 0.3995 - val_loss: 1.6984
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.4423 - loss: 1.5692
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.4286 - loss: 1.6426
SGD Accuracy: 0.4422999918460846
Adam Accuracy: 0.428600013256073
```

